

Benchmarking AI Vector Databases for Genesis Mission

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Overview

In support of the **Genesis Mission**, the U.S. government's national AI initiative, **Circe** is an HPC-hosted AI service environment that supports model access, custom knowledge databases, and external tool integration [1]. As shown in Fig. 1, a central feature of Circe is effective **retrieval-augmented generation (RAG)**, requiring streamlined vector databases. We benchmarked five databases — **Qdrant**, **Redis**, **Weaviate**, **ChromaDB**, and **Elasticsearch** — and compared their scalability, query speed, latency, and retrieval accuracy.

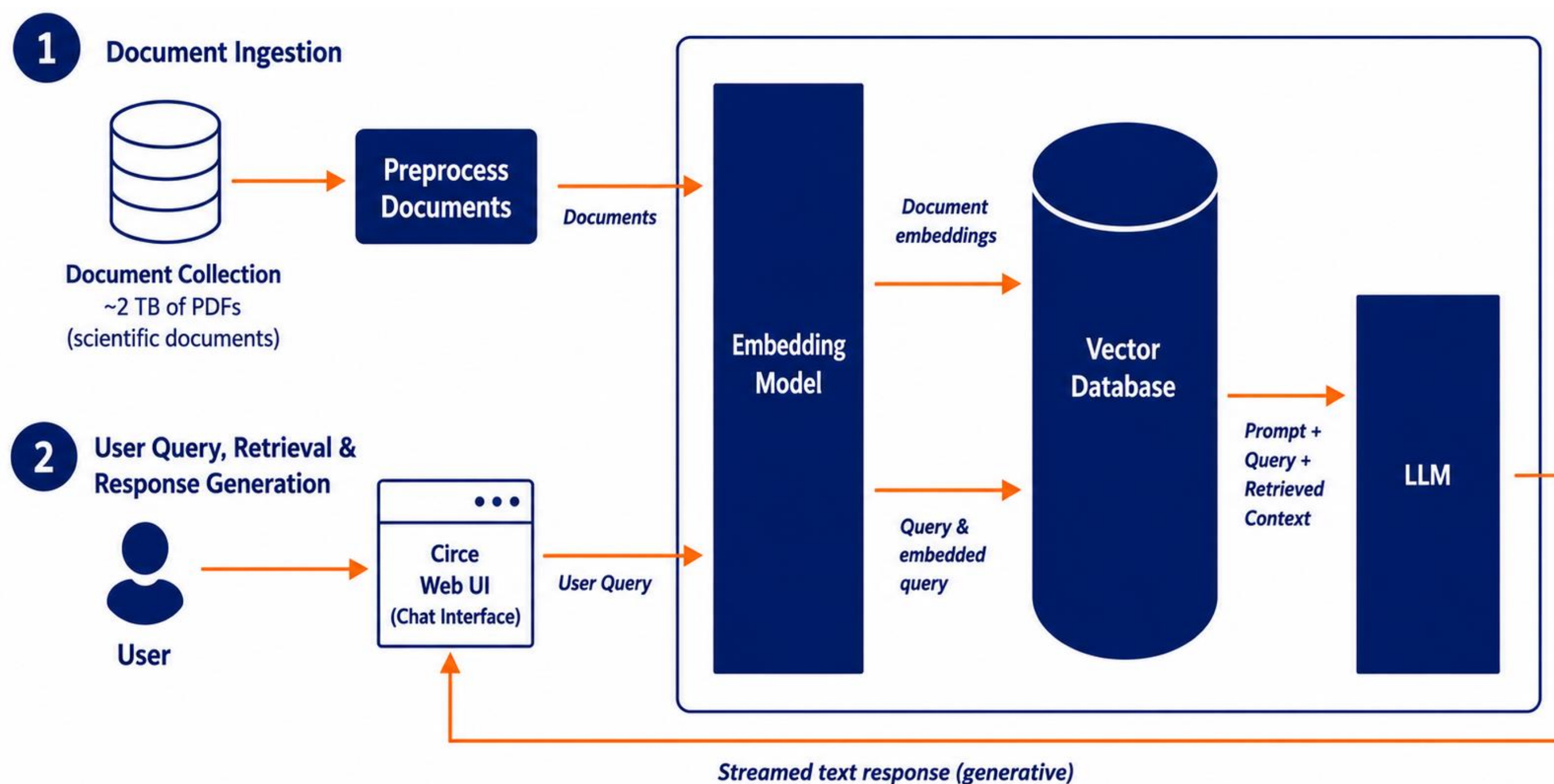


Figure 1. Circe RAG (Retrieval-Augmented Generation) workflow

Methods

TEST SETUP & ENVIRONMENT

- **Hardware:** Skylake Platinum partition (Intel Xeon Platinum CPUs) on Darwin testbed
- **Tool:** Qdrant's vector-db-benchmark framework [3]
- **Dataset:** 2.7 million 4096-dimensional embeddings vectors representative of target PDF-derived data from arXiv [4]
- **Configuration:** shared HNSW setup with database and benchmark tools pinned to separate CPU sockets
 - Hierarchical Navigable Small World: graph-based approximate nearest neighbor search
- **Performance Metrics:** throughput, mean latency, 95th/99th percentile latency (p95/p99), total search time, and retrieval precision [2]

BENCHMARK TRIALS

- **Baseline:** default HNSW configuration (M = 32, ef_construct = 256, search_ef = 128, and parallel_connections = 64) with no optimizations
- **Compressed:** float 32-bit to integer 8-bit quantization for speed and memory gains
- **Configuration-tuned:** best performing configuration for each database found during testing

Results

Three benchmark configurations were tested: baseline, compressed, and configuration-tuned. Elasticsearch failed to complete any benchmark runs, while ChromaDB only completed the baseline test due to no native vector quantization support. The highlighted green region represents **optimal performance**.

LATENCY AND PRECISION

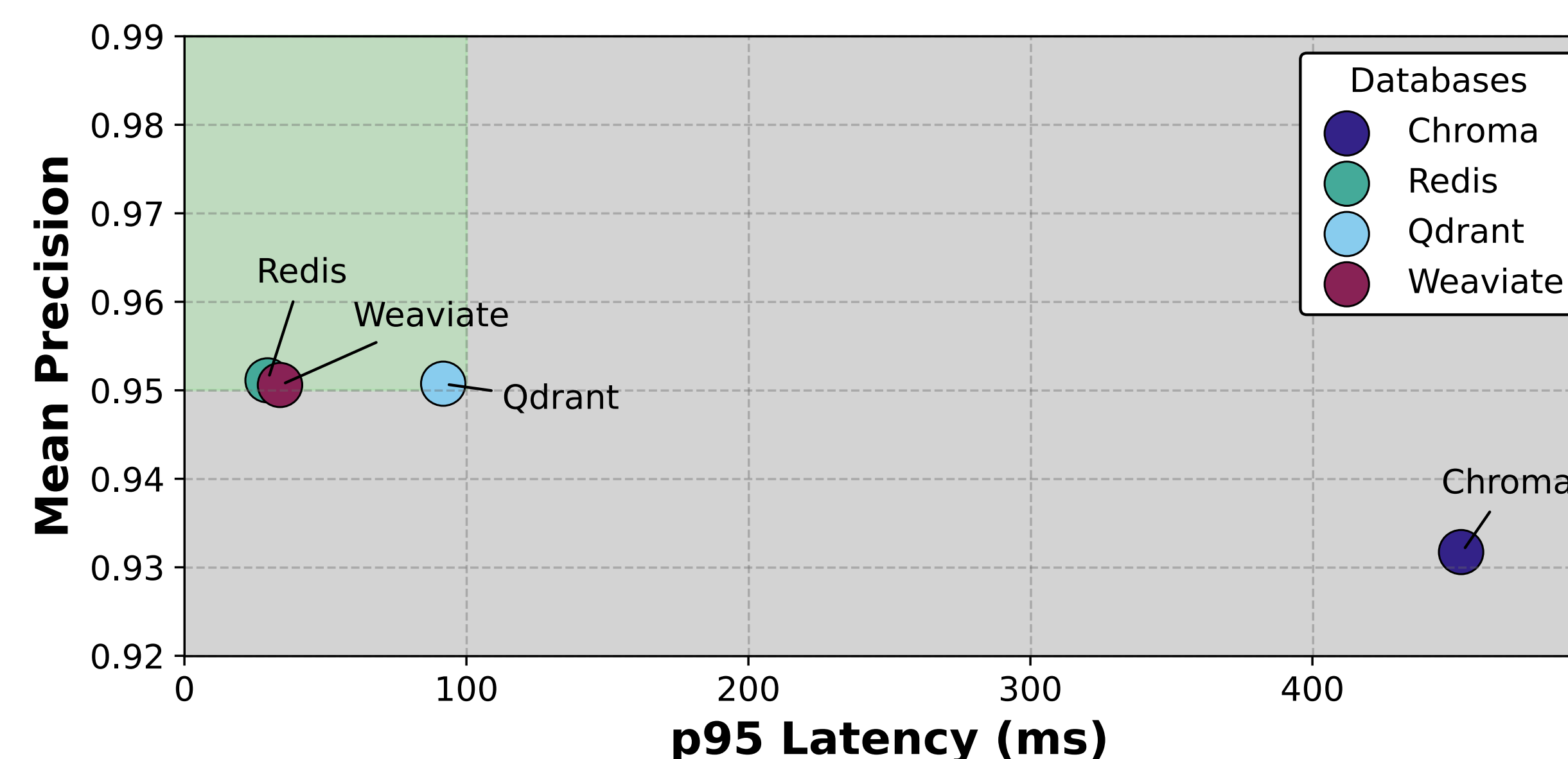


Figure 2. Baseline performance: shared HNSW configuration

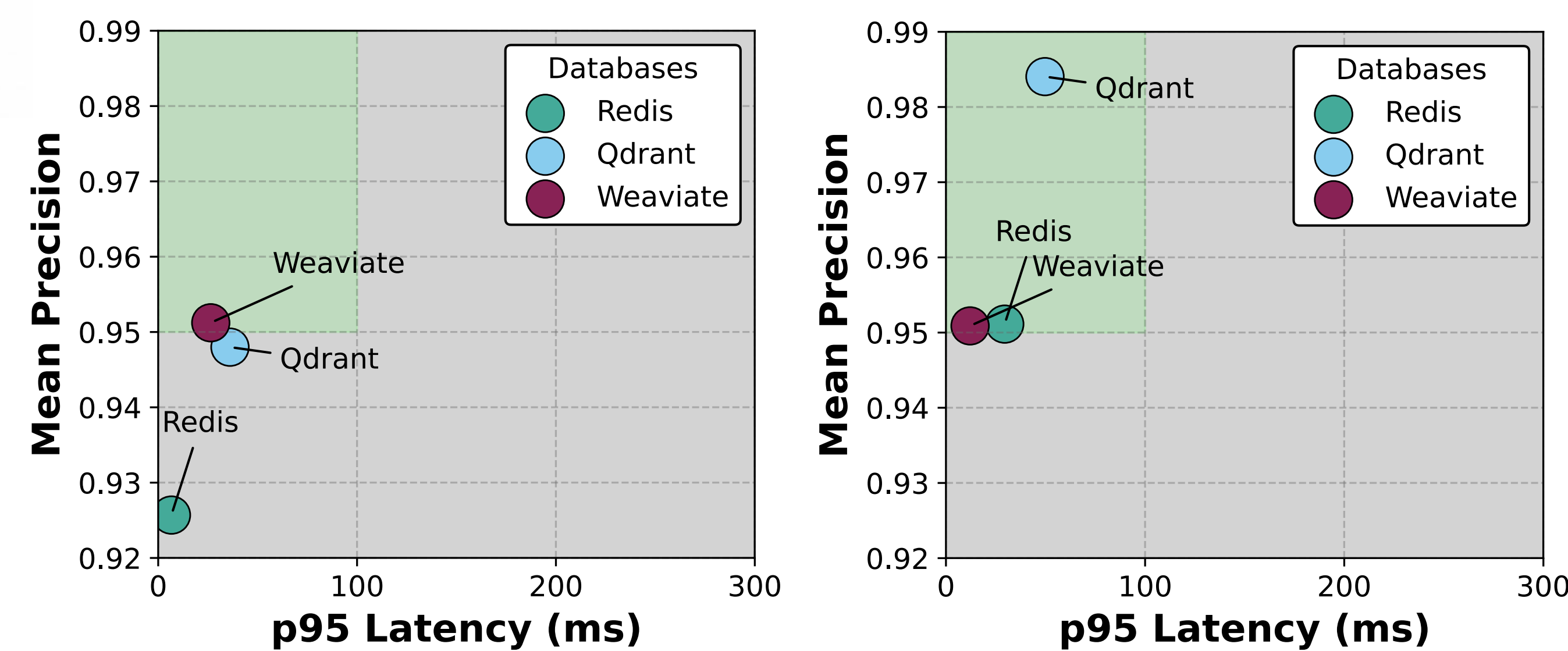


Figure 3. Compressed performance: scalar int8 quantization, no rescoring

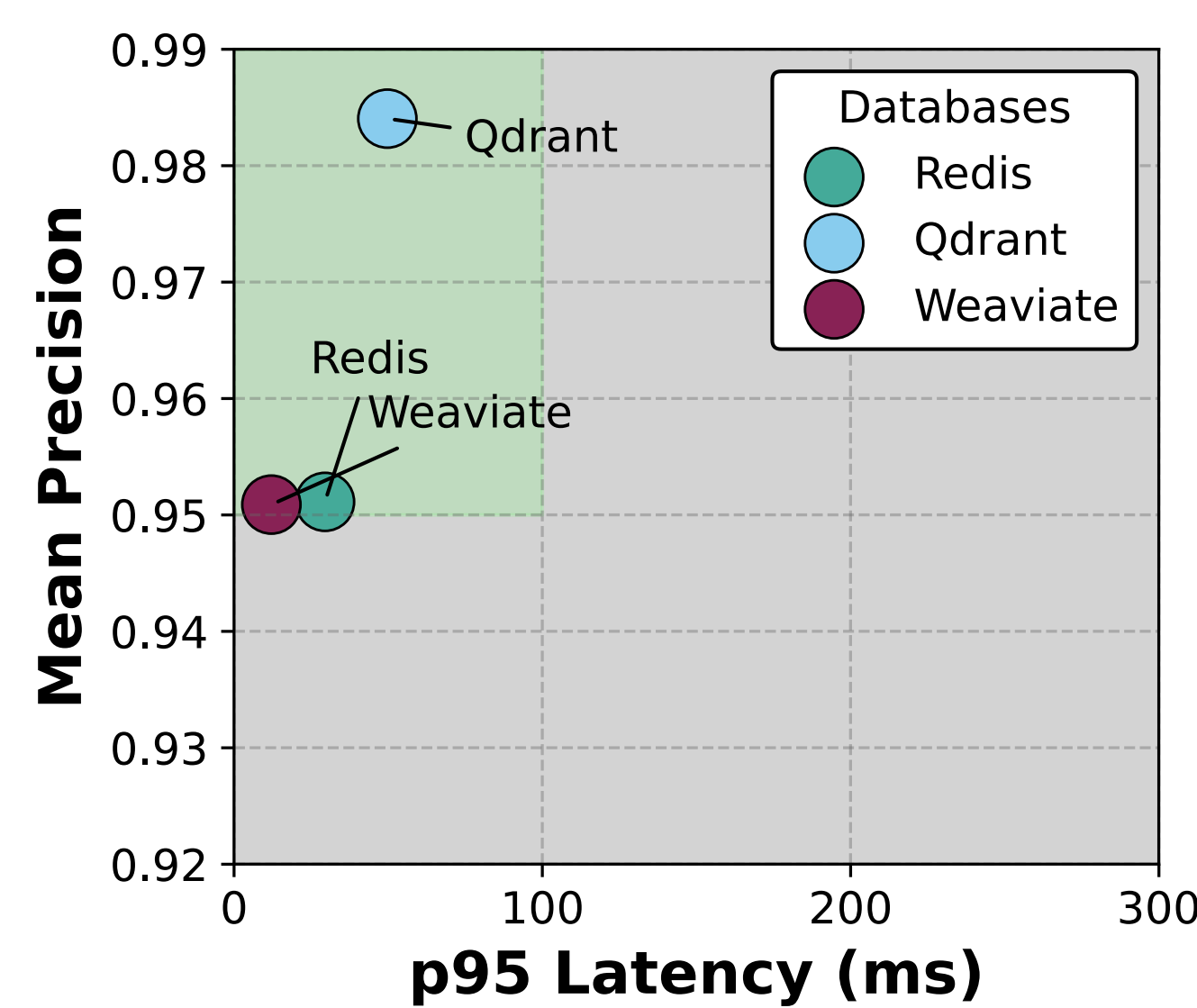


Figure 4. Configuration-tuned performance: best database-specific configuration

THROUGHPUT

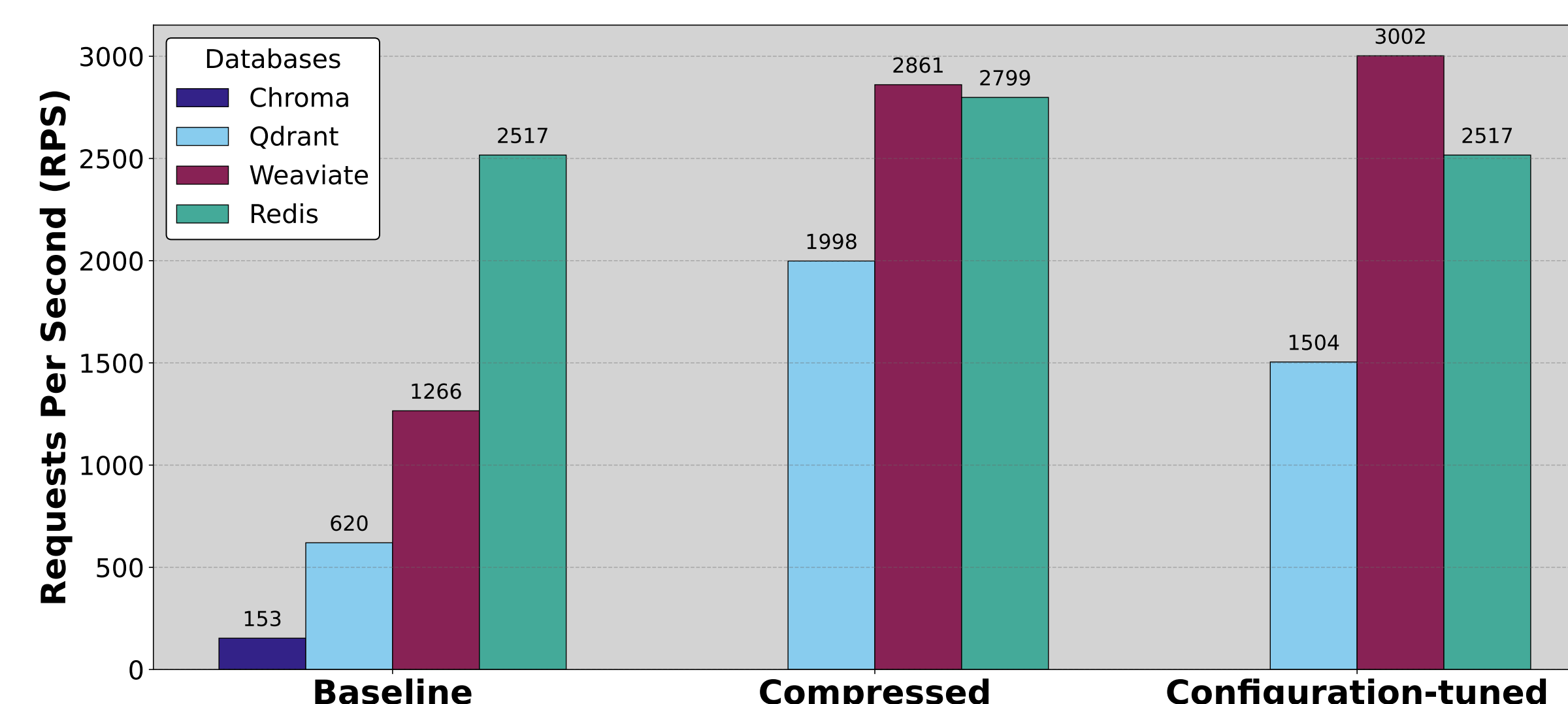


Figure 5. Requests per second (RPS) across each benchmark run

Database	Mean Precision	p95 (ms)	p99 (ms)	RPS
Redis	0.9511	29.5	30.9	2516.7
Qdrant	0.9507	91.8	95.5	620.0
Weaviate	0.9506	33.9	41.1	1265.7
ChromaDB	0.9317	452.7	460.5	153.3

Table 1. Database Baseline Evaluation

Database	Mean Precision	p95 (ms)	p99 (ms)	RPS
Redis	0.9511	29.5	30.9	2516.7
Qdrant	0.9840	49.7	51.7	1504.3
Weaviate	0.9509	12.1	15.4	3002.3

Table 2. Database Configuration-tuned Evaluation

Analysis

BASELINE PERFORMANCE (FIG. 2)

- **Redis achieved the strongest baseline performance.**
- **Weaviate ranked second** at 1,265.7 RPS, followed by **Qdrant** at 620.0 RPS.
- **ChromaDB had the weakest baseline performance**, reaching 153.3 RPS and recording the highest latency.

CONFIGURATION-TUNED PERFORMANCE (FIG. 4 & 5)

- **Weaviate achieved the highest optimized throughput**, increasing from 1,265.7 to 3,002.3 RPS and its p95 latency decreasing from 33.9 to 12.1 ms
- **Qdrant demonstrated the largest proportional throughput improvement**, increasing from 620.0 to 1,504.3 RPS
- Qdrant's p95 and p99 latency both decreased by approximately **46%** and it also achieved the **highest measured precision of 0.9840**.

Conclusion

Vector database selection should balance throughput, latency, precision, and scalability. No single database led every metric: **Weaviate is the strongest candidate for maximum query speed**, while **Qdrant offers the best balance of tunability and retrieval accuracy (tab. 2)**. Future work should evaluate long-running performance under realistic concurrent RAG workloads.

References

- [1] Los Alamos National Laboratory, High Performance Computing Division, Circe: AI Services Platform, internal computing service, Los Alamos, NM, USA, 2026.
- [2] H. Simhadri et al., "Results of the Big ANN: NeurIPS'23 competition," 2024, arXiv:2409.17424.
- [3] "qdrant/vector-db-benchmark: Framework for benchmarking vector search engines," GitHub, 2026. <https://github.com/qdrant/vector-db-benchmark>.
- [4] Scalable Parallel Computing Laboratory (SPCL), "arxiv-for-fanns-large," huggingface, 2026. <https://huggingface.co/datasets/spcl/arxiv-for-fanns-large>.